Solutions to the exercises in *Categorical Logic and Type Theory* by Bart Jacobs

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Chapter 1

Introduction to fibred category theory

1.1 Fibrations

1.1.1.

(i) Prove Proposition 1.1.4.[a]

(ii) Suppose f is Cartesian and g and h are above the same map. Show that $f \circ g = f \circ h$ implies $g = h$.

[a] [2, Proposition 1.1.4] asserts that Cartesian liftings are unique up-to-isomorphism (in a slice): if f and f' with $\operatorname{cod} f = \operatorname{cod} f'$ are both Cartesian over the same map, then there is a unique vertical isomorphism $\varphi : X \xrightarrow{\cong} X'$ with $f' \circ \varphi = f$.

We will prove this in the more general context of displayed categories (cf. [1]).

Solution, (i). Suppose $\mathbb{E}$ is a displayed category over $\mathbb{B}$, $u : I \to J$ is a morphism in $\mathbb{B}$, and $f : X \to Y$ and $f' : X' \to Y$ are Cartesian morphisms in $\mathbb{E}$ over u. Since f' is Cartesian and f is a morphism into Y over $u = u \circ \mathbf{1}_I$, there exists a unique morphism $\varphi : X \to X'$ in $\mathbb{E}$ over I such that $f = f' \circ \varphi$. This is summarized by the following diagram.

$$
\begin{tikzcd}
X \arrow[d] \arrow[rr, bend left, "f"] & & \\
& X' \arrow[r, "f'"] \arrow[d] \arrow[dr, bend left, "u"] & Y \arrow[d] \\
I \arrow[r, equal] & I \arrow[r, "u"] & J
\end{tikzcd}
\quad \Longrightarrow \quad
\begin{tikzcd}
X \arrow[d] \arrow[rr, bend left, "f"] \arrow[dr, dashed, "\exists!\,\varphi"] & & \\
& X' \arrow[r, "f'"] \arrow[d] & Y \arrow[d] \\
I \arrow[r, equal] & I \arrow[r, "u"] & J
\end{tikzcd}
$$

Similarly, there exists a unique morphism $\psi : X' \to X$ in $\mathbb{E}$ over I such that $f' = f \circ \psi$. Moreover, both $\mathbf{1}_{X'}$ and $\varphi \circ \psi$ are morphisms in $\mathbb{E}$ over I whose

composition with f' is f' itself, as suggested in the following diagram.

$$\begin{tikzcd}
X' \arrow[dr, "\psi"] \arrow[drr, bend left, "f'"] \arrow[ddr, bend right, "\mathbf{1}_{X'}"'] \arrow[dd] & & \\
& X \arrow[dr, "\varphi"] \arrow[drr, bend left, "f"] \arrow[d] & & \\
I \arrow[dr, equal] & I \arrow[dr, equal] & X' \arrow[r, "f'"] \arrow[d] & Y \arrow[d] \\
& & I \arrow[r, "u"] & J
\end{tikzcd}$$

Thus, since f' is Cartesian, $\varphi \circ \psi = \mathbf{1}_{X'}$. By symmetry, $\psi \circ \varphi = \mathbf{1}_X$, and hence φ is an isomorphism. $\square$

Solution, (ii). Suppose $\mathbb{E}$ is a displayed category over $\mathbb{B}$, $u : I \to J$ and $v : K \to I$ are morphisms in $\mathbb{B}$, $f : X \to Y$ is a Cartesian morphism in $\mathbb{E}$ over u, and $g, h : Z \to X$ are morphisms in $\mathbb{E}$ over v such that $f \circ g = f \circ h$. These suppositions are summarized in the following commutative diagram.

$$\begin{tikzcd}
Z \arrow[dr, bend left, "g"] \arrow[drr, bend left, "f\circ g = f\circ h"] \arrow[dr, bend right, "h"'] \arrow[dd] & & \\
& X \arrow[r, "f"] \arrow[d] & Y \arrow[d] \\
K \arrow[dr, "v"'] \arrow[urr, bend left] & I \arrow[r, "u"] & J \\
& & \arrow[ul, bend left, "u\circ v"]
\end{tikzcd}$$

Since f is Cartesian, this situation immediately implies that $g = h$. $\square$

1.1.2. Let $p : \mathbb{E} \to \mathbb{B}$ be a functor; assume $f : X \to Y$ is in $\mathbb{E}$ and put $u = pf$. Show that f is Cartesian if and only if for each $Z \in \mathbb{E}$ and $v : pZ \to pX$ in $\mathbb{B}$, the function

$$\mathbb{E}_v(Z, X) \xrightarrow{f\circ(-)} \mathbb{E}_{u\circ v}(Z, Y)$$

is an isomorphism.

We will prove this in the more general context of displayed categories (cf. [1]).

Solution. Suppose $\mathbb{E}$ is a displayed category over $\mathbb{B}$, $u : I \to J$ is a morphism in $\mathbb{B}$, and $f : X \to Y$ is a morphism in $\mathbb{E}$ over u.

($\Longrightarrow$) Suppose f is Cartesian, and take a morphism $v : K \to I$ in $\mathbb{B}$ and an object Z in $\mathbb{E}$ over K. Since f is Cartesian, this situation gives rise to a one-to-one correspondence between the dashed arrows in the diagram below.

$$\begin{tikzcd}
Z \arrow[dd] \arrow[dr, dashed] \arrow[drr, dashed, bend left] & & \\
& X \arrow[r, "f"] \arrow[dd] & Y \arrow[dd] \\
K \arrow[dr, "v"'] \arrow[drr, bend left, "u\circ v"] & & \\
& I \arrow[r, "u"] & J
\end{tikzcd}$$

Indeed, given a morphism $g : Z \to X$ in $\mathbb{E}$ over v, composition with f yields a morphism $f \circ g : Z \to Y$ over $u \circ v$. Conversely, given a morphism $h : Z \to Y$ in $\mathbb{E}$ over $u \circ v$, there is a unique morphism $g : Z \to X$ in $\mathbb{E}$ over v such that $f \circ g = h$ since f is Cartesian. This is precisely the assertion that $\mathbb{E}_v(Z, X) \xrightarrow{f\circ(-)} \mathbb{E}_{u\circ v}(Z, Y)$ is an isomorphism (of sets).

($\Longleftarrow$) Conversely, suppose that $\mathbb{E}_v(Z, X) \xrightarrow{f\circ(-)} \mathbb{E}_{u\circ v}(Z, Y)$ is an isomorphism for every morphism $v : K \to I$ in $\mathbb{B}$ and an object Z in $\mathbb{E}$ over K. Consider a particular morphism $v : K \to I$ in $\mathbb{B}$ and an object Z in $\mathbb{E}$ over K, and suppose there is a morphism $h : Z \to Y$ in $\mathbb{E}$ over $u \circ v$. Then the assumed isomorphism implies that there is a unique morphism $g : Z \to X$ in $\mathbb{E}$ over v such that $h = f \circ g$, which shows that f is Cartesian. $\square$

1.1.3. Consider the total category of a fibration. Show that

(i) every morphism factors as a vertical map followed (diagrammatically) by a Cartesian one;

(ii) a Cartesian map above an isomorphism is an isomorphism. Especially a vertical Cartesian map is an isomorphism.

Solution, (i). Let $\mathbb{E}$ be a fibration over $\mathbb{B}$, and let $f : X \to Y$ be a morphism in $\mathbb{E}$ over a morphism $u : I \to J$ in $\mathbb{B}$. Then there exists a Cartesian morphism $f' : X' \to Y$ in $\mathbb{E}$ above u. Then we are in the situation of the diagram on the left below, so there exists a unique (vertical) morphism $g : X \to X'$ over I such that the diagram below on the right commutes.

$$\begin{tikzcd}
X \arrow[d] \arrow[rrr, bend left, "f"] & & & \\
& X' \arrow[r, "f'"] \arrow[d] & Y \arrow[d] & \\
I \arrow[urr, bend left] \arrow[r, equal] & I \arrow[r, "u"] \arrow[urr, bend left, "u"] & J &
\end{tikzcd}
\Longrightarrow
\begin{tikzcd}
X \arrow[d] \arrow[dr, dashed, "g"] \arrow[rrr, bend left, "f"] & & & \\
& X' \arrow[r, "f'"] \arrow[d] & Y \arrow[d] & \\
I \arrow[urr, bend left] \arrow[r, equal] & I \arrow[r, "u"] \arrow[urr, bend left, "u"] & J &
\end{tikzcd}$$

In particular, $f = f' \circ g$; in other words, f factors as a vertical morphism followed by a Cartesian morphism. $\square$

Solution, (ii). Let $\mathbb{E}$ be a fibration over $\mathbb{B}$, and let $f : X \to Y$ be a Cartesian morphism in $\mathbb{E}$ over an isomorphism $u : I \to J$ in $\mathbb{B}$. Let $v = u^{-1} : J \to I$ in $\mathbb{B}$, and consider the situation in the following commutative diagram.

$$\begin{tikzcd}
Y \arrow[d] \arrow[dr, dashed, "\exists! g"] \arrow[rrr, bend left, "\mathbf{1}_Y"] & & & \\
& X \arrow[r, "f"] \arrow[d] & Y \arrow[d] & \\
J \arrow[r, "\sim", "v"'] \arrow[urr, bend left, "\mathbf{1}_J"] & I \arrow[r, "\sim", "u"'] & J &
\end{tikzcd}$$

Here the dashed morphism $g : Y \to X$ exists uniquely by virtue of f being Cartesian. Since $f \circ g = \mathbf{1}_Y$ is Cartesian (by Exercise 1.1.4(i)) and f is Cartesian, Exercise 1.1.4(iii) implies that g is Cartesian over the isomorphism v, and $f \circ g = \mathbf{1}_Y$. The same argument then implies that there exists a morphism $h : X \to Y$ over $u = v^{-1}$ such that $g \circ h = \mathbf{1}_X$. But then

$$f = f \circ \mathbf{1}_X = f \circ g \circ h = \mathbf{1}_Y \circ h = h,$$

so $g \circ f = \mathbf{1}_X$, and hence $g = f^{-1}$, so f is an isomorphism. $\square$

1.1.4. Let $\begin{array}{c}\mathbb{E}\\ \downarrow p\\ \mathbb{B}\end{array}$ be a fibration. Prove that

(i) all isomorphisms in $\mathbb{E}$ are Cartesian;

(ii) if $\xrightarrow{f}\xrightarrow{g}$ is a composable pair of Cartesian morphisms in $\mathbb{E}$, then also their composite $\xrightarrow{g\circ f}$ is Cartesian.

Hence it makes sense to talk about the subcategory $\mathrm{Cart}(\mathbb{E}) \hookrightarrow \mathbb{E}$ having all objects from $\mathbb{E}$ but only the Cartesian arrows. We write $|p|$ for the composite $\mathrm{Cart}(\mathbb{E}) \hookrightarrow \mathbb{E} \to \mathbb{B}$.

(iii) Let $\xrightarrow{f}\xrightarrow{g}$ be a composable pair in $\mathbb{E}$ again. Show now that if g and $g \circ f$ are Cartesian, then f is Cartesian as well.

(iv) Verify that a consequence of (iii) is that the functor $|p| : \mathrm{Cart}(\mathbb{E}) \to \mathbb{B}$ is a fibration. From (ii) in the previous exercise it follows that all fibres of $|p|$ are groupoids (*i.e.* that all maps in the fibres are isomorphisms).

[This $|p|$ will be called the **fibration of objects** of p.]

We will prove this in the more general context of displayed categories (cf. [1]).

Solution, (i). Let $\mathbb{E}$ be a displayed category over $\mathbb{B}$ (not necessarily a fibration), and let $f : X \to Y$ be a displayed isomorphism in $\mathbb{E}$ above an isomorphism $u : I \to J$ in $\mathbb{B}$. Suppose $v : K \to I$ is a morphism in $\mathbb{B}$ and $g : Z \to Y$ is a morphism in $\mathbb{E}$ above $u \circ v : K \to J$. Then $f^{-1} \circ g : Z \to X$ is the unique morphism in $\mathbb{E}$ above v such that the following diagram commutes.

$$\begin{tikzcd}
Z \arrow[dr, dashed, "f^{-1}\circ g"] \arrow[drr, bend left, "g"] \arrow[dd] & & \\
& X \arrow[r, "f"] \arrow[dd] & Y \arrow[dd] \\
K \arrow[dr, "v"'] \arrow[drr, bend left, "u\circ v"] & & \\
& I \arrow[r, "u"] & J
\end{tikzcd}$$

Indeed, to verify uniqueness, suppose $h : Z \to X$ is another morphism in $\mathbb{E}$ above v such that $f \circ h = g$, and observe that

$$h = f^{-1} \circ f \circ h = f^{-1} \circ g.$$

Therefore f is Cartesian. $\square$

Solution, (ii). Suppose $\mathbb{E}$ is a displayed category over $\mathbb{B}$ (not necessarily a fibration), $u : I \to J$ and $v : J \to K$ are morphisms in $\mathbb{B}$, and $f : X \to Y$ and $g : Y \to Z$ are Cartesian morphisms in $\mathbb{E}$ over u and v, respectively. To show that $g \circ f : X \to Z$ is Cartesian over $v \circ u : I \to K$, consider a morphism $w : L \to I$ in $\mathbb{B}$ and a morphism $h : W \to Z$ in $\mathbb{E}$ above $v \circ u \circ w$. These data are summarized in the following commutative diagram.

$$\begin{tikzcd}
W \arrow[d] \arrow[rrr, bend left, "h"] & & & \\
 & X \arrow[r, "f"] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & Y \arrow[r, "g"] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & Z \arrow[d] \\
L \arrow[dr, "w"] & I \arrow[r, "u"] & J \arrow[r, "v"] & K \\
 & & &
\end{tikzcd}$$

Since g is Cartesian, there exists a unique morphism $h' : W \to Y$ in $\mathbb{E}$ above $u \circ w$ such that $h = g \circ h'$, as in the following commutative diagram.

$$\begin{tikzcd}
W \arrow[d] \arrow[rrr, bend left, "h"] \arrow[rr, dashed, bend left, "h'"] & & & \\
 & X \arrow[r, "f"] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & Y \arrow[r, "g"] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & Z \arrow[d] \\
L \arrow[dr, "w"] & I \arrow[r, "u"] & J \arrow[r, "v"] & K \\
 & & &
\end{tikzcd}$$

Moreover, since f is Cartesian, there exists a unique morphism $h'' : W \to X$

in $\mathbb{E}$ above w such that $h' = f \circ h''$, as in the following commutative diagram.

$$\begin{tikzcd}
W \arrow[dr, dashed, "h''"] \arrow[drr, bend left, "h'"] \arrow[drrr, bend left, "h"] \arrow[dd] & & & \\
& X \arrow[r, "f"] \arrow[dd] \arrow[dr, phantom, "\lrcorner"] & Y \arrow[r, "g"] \arrow[dd] \arrow[dr, phantom, "\lrcorner"] & Z \arrow[dd] \\
L \arrow[dr, "w"] & & & \\
& I \arrow[r, "u"] & J \arrow[r, "v"] & K
\end{tikzcd}$$

In particular, we have

$$h = g \circ h' = g \circ f \circ h''.$$

It remains to show that if $i : W \to X$ is another morphism in $\mathbb{E}$ above w such that $h = g \circ f \circ i$, then $i = h''$. Indeed, in this case the uniqueness of h' implies that $f \circ i = h'$, and so the uniqueness of h'' implies that $i = h''$. Thus, $g \circ f$ is Cartesian. $\square$

Solution, (iii). Suppose $\mathbb{E}$ is a displayed category over $\mathbb{B}$ (not necessarily a fibration), $u : I \to J$ and $v : J \to K$ are morphisms in $\mathbb{B}$, and $f : X \to Y$ and $g : Y \to Z$ are morphisms in $\mathbb{E}$ over u and v, respectively, such that g and $g \circ f$ are Cartesian. To show that $f : X \to Y$ is Cartesian over u, consider a morphism $w : L \to I$ in $\mathbb{B}$ and a morphism $h : W \to Y$ in $\mathbb{E}$ above $u \circ w$. This situation is summarized in the following commutative diagram.

$$\begin{tikzcd}
W \arrow[drrr, bend left, "h"] \arrow[dd] & & & \\
& X \arrow[r, "f"] \arrow[dd] & Y \arrow[r, "g"] \arrow[dd] & Z \arrow[dd] \\
L \arrow[dr, "w"] & & & \\
& I \arrow[r, "u"] & J \arrow[r, "v"] & K
\end{tikzcd}$$

But then $g \circ h : W \to Z$ is a morphism in $\mathbb{E}$ above $v \circ u \circ w$, so since $g \circ f$ is Cartesian, there exists a unique morphism $h' : W \to X$ in $\mathbb{E}$ over w such that $g \circ h = g \circ f \circ h'$. Since g is Cartesian, Exercise 1.1.1(i) implies that $h = f \circ h'$. Moreover, suppose $i : W \to X$ is another morphism above w such that $h = f \circ i$. Then $g \circ h = g \circ f \circ i$, so the uniqueness of h' arising from $g \circ f$ being Cartesian implies that $i = h'$. Thus, f is Cartesian. $\square$

Solution, (iv). Let $p : \mathbb{E} \to \mathbb{B}$ be a fibration; we will show that $|p| : \mathrm{Cart}(\mathbb{E}) \to \mathbb{B}$ is also a fibration. Consider a morphism $u : I \to J$ in $\mathbb{B}$ and an object Y in $\mathbb{E}$ over J. Since p is a fibration, there exists a p-Cartesian morphism $f : X \to Y$ in $\mathbb{E}$ over u. We claim that f is $|p|$-Cartesian as well. Indeed, suppose there is a morphism $v : K \to I$ in $\mathbb{B}$ and a p-Cartesian morphism $h : Z \to Y$ in $\mathbb{E}$ over $u \circ v$. Since f is p-Cartesian, there exists a unique morphism $h : Z \to X$ in $\mathbb{E}$ over v such that $g = f \circ h$. But now (iii) implies that h is p-Cartesian as well, so the entire diagram below is actually a diagram in the displayed category induced by $|p| : \mathrm{Cart}(\mathbb{E}) \to \mathbb{B}$.

$$\begin{tikzcd}
Z \arrow[dr, dashed, "\exists! h"] \arrow[drr, bend left, "g"] \arrow[dd] & & \\
& X \arrow[r, "f"] \arrow[dd] & Y \arrow[dd] \\
K \arrow[dr, "v"'] \arrow[drr, bend left, "u \circ v"] & & \\
& I \arrow[r, "u"] & J
\end{tikzcd}$$

Therefore, f is $|p|$-Cartesian, so $|p|$ is a fibration. $\square$

1.1.5. Verify that the following two results—known as the Pullback Lemmas—are a consequence of (ii) and (iii) in the previous exercise. Consider

$$\begin{tikzcd}
\bullet \arrow[r] \arrow[d] \arrow[dr, phantom, "(B)"] & \bullet \arrow[r] \arrow[d] \arrow[dr, phantom, "(A)"] & \bullet \arrow[d] \\
\bullet \arrow[r] & \bullet \arrow[r] & \bullet
\end{tikzcd}$$

(i) If (A) and (B) are pullback squares, then the outer rectangle is also a pullback square.

(ii) If the outer rectangle and (A) are pullback squares, then (B) is a pullback square as well.

Solution. Recall that in a category $\mathbb{B}$, the assertion that the commutative

square

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "p"] & Y \arrow[d, "q"] \\ I \arrow[r, "u"] & J \end{tikzcd}$$

is a pullback square if and only if (u, f) is a Cartesian lift of u in the context of the codomain functor $\mathrm{cod} : \mathbb{B}^{\to} \to \mathbb{B}$, as in the following diagram.

$$\begin{tikzcd} \begin{pmatrix} X \\ \downarrow p \\ I \end{pmatrix} \arrow[r, "(u,f)"] \arrow[d, "\mathrm{cod}"] & \begin{pmatrix} Y \\ \downarrow q \\ J \end{pmatrix} \arrow[d, "\mathrm{cod}"] \\ I \arrow[r, "u"] & J \end{tikzcd}$$

Therefore, (i) and (ii) follow immediately from Exercise 1.1.4(ii) and Exercise 1.1.4(iii), respectively. □

1.1.6. Consider a functor $p : \mathbb{E} \to \mathbb{B}$. We describe a slightly weaker notion of Cartesianness, than the one above. Call a morphism $f : X \to Y$ in $\mathbb{E}$ **weak Cartesian** if for each $g : Z \to Y$ with $pf = pg$ there is a unique vertical $h : Z \to X$ with $f \circ h = g$. Show that the functor p is a fibration if and only if both

(a) every morphism $u : I \to pY$ in $\mathbb{B}$ has a weak Cartesian lifting $f : X \to Y$;

(b) the composition of two weak Cartesian morphisms is again weak Cartesian.

Solution. ($\Longrightarrow$) Since every Cartesian morphism is weak Cartesian and compositions of Cartesian morphisms are Cartesian by Exercise 1.1.4(ii), every fibration satisfies (a) and (b).

($\Longrightarrow$) Conversely, suppose (a) and (b) hold. Let $u : I \to J$ be a morphism in $\mathbb{B}$, and let Y be an object in $\mathbb{E}$ above J. By (a), let $f : X \to Y$ be a weak Cartesian lift of u in $\mathbb{E}$. Consider another morphism $v : K \to I$ in $\mathbb{B}$ and a

morphism $g : Z \to Y$ in $\mathbb{E}$ over $u \circ v$. By (a), choose a weak Cartesian lift $p : W \to X$ in $\mathbb{E}$ of v. Now $f \circ p$ is weak Cartesian over $u \circ v$ by (b), so there exists a unique vertical morphism $q : Z \to W$ over K such that the following diagram commutes.

$$\begin{tikzcd}
Z \arrow[dr, dashed, "q"] \arrow[drrr, bend left, "g"] \arrow[dd] & & & \\
& W \arrow[r, "p"] \arrow[dd] & X \arrow[r, "f"] \arrow[dd] & Y \arrow[dd] \\
K \arrow[dr, equal] & & & \\
& K \arrow[r, "v"] & I \arrow[r, "u"] & J
\end{tikzcd}$$

In particular, the following diagram commutes.

$$\begin{tikzcd}
Z \arrow[dr, "p \circ q"] \arrow[drrr, bend left, "g"] \arrow[dd] & & & \\
& X \arrow[r, "f"] \arrow[dd] & Y \arrow[dd] \\
K \arrow[dr, "v"] & & & \\
& I \arrow[r, "u"] & J
\end{tikzcd}$$

It remains to show that if $h : Z \to X$ is another morphism in $\mathbb{E}$ above v such that $g = f \circ h$, then $h = p \circ q$. In this case, since p is weak Cartesian, we can find a unique vertical morphism $q' : Z \to W$ over K such that $h = p \circ q'$.

$$\begin{tikzcd}
Z \arrow[dr, dashed, "q'"] \arrow[drr, bend left, "h"] \arrow[drrr, bend left, "g"] \arrow[dd] & & & \\
& W \arrow[r, "p"] \arrow[dd] & X \arrow[r, "f"] \arrow[dd] & Y \arrow[dd] \\
K \arrow[dr, equal] & & & \\
& K \arrow[r, "v"] & I \arrow[r, "u"] & J
\end{tikzcd}$$

In particular, $g = f \circ p \circ q'$, so the uniqueness of q implies that $q' = q$, and hence $h = p \circ q$. Thus, f is a Cartesian lift of u, so p is a fibration. $\square$

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